Crowd Density Estimation with CSRNet

Experimental results on ShanghaiTech Part A, Part B and the Roboflow crowd dataset

Model: CSRNet (VGG-16 front-end, dilated back-end), checkpoint epoch 25 | Report generated 16 October 2026

77.3	19.1	48.5	78.9%	76.9%
MAE ShanghaiTech A	MAE ShanghaiTech B	MAE Roboflow	Density-level accuracy	Macro F1

1. Summary

A single CSRNet model was trained jointly on ShanghaiTech Part A, ShanghaiTech Part B, the Roboflow crowd dataset and 100 pseudo-labelled phone photographs, and was evaluated on 956 annotated test images it never trained on. It reaches an MAE of 77.3 on ShanghaiTech Part A, 19.1 on Part B and 48.5 on the Roboflow test split, with R^2 = 0.881 between predicted and true counts over all test images. When counts are grouped into four crowd-density levels (Low, Medium, High, Very high), the predicted level matches the true level for 78.9% of the images (macro F1 76.9%, quadratic-weighted Cohen's kappa 0.842) and is within one level for 99.9%.

2. Experimental setup

2.1 Datasets

ShanghaiTech Part A contains dense crowd images collected from the internet; Part B contains sparser street scenes from Shanghai at a fixed 1024x768 resolution. Both are annotated with one point per head. The Roboflow crowd dataset (640x640 images) is annotated with one bounding box per head, whose centre is used as the head point. Ground-truth density maps use geometry-adaptive Gaussian kernels (Part A, Roboflow) and a fixed kernel (Part B), and each map sums to the annotated head count. The 100 phone photographs have no annotations; they were pseudo-labelled by the Part A model and used for training only, never for evaluation.

Table 1. Datasets and splits. Count statistics are computed on the test split.

Dataset	Train	Validation	Test	Test count range	Test mean count
ShanghaiTech Part A	300	-	182	66 - 2256	433.3
ShanghaiTech Part B	400	-	316	9 - 539	123.8
Roboflow crowd	3212	917	458	10 - 1202	182.3
Phone photos (pseudo-labelled)	100	-	-	-	-

2.2 Model and training

CSRNet uses the first ten convolutional layers of VGG-16 as a front-end and six dilated 3x3 convolutions (dilation 2) as a back-end, followed by a 1x1 convolution that outputs a density map at 1/8 of the input resolution; the predicted count is the sum of the map. The network was first trained on ShanghaiTech Part A from ImageNet weights, then fine-tuned jointly on all training sets.

Table 2. Training configuration of the evaluated model.

Setting	Value
Initialisation	ShanghaiTech Part A best checkpoint
Optimiser	Adam, learning rate 1e-5, weight decay 1e-4
Loss	Pixel-wise MSE between density maps (sum reduction)
Batch / crop	8 images, random 256x256 crops, random horizontal flip
Precision	Mixed precision (AMP) on an NVIDIA T4 GPU (Google Colab)
Epochs	25 completed; best validation MAE at epoch 25
Time per epoch	160 s (mean)
Model selection	Lowest MAE on ShanghaiTech A + B test and Roboflow valid

2.3 Evaluation metrics

For N test images with true count y and predicted count p: MAE = mean $|p - y|$ and RMSE = sqrt(mean $(p - y)^2$) (reported as "MSE" in most crowd-counting papers). Bias is the mean signed error (positive = over-counting). Counting accuracy is mean max(0, 1 - $|p - y| / y$). The 95% confidence interval of the MAE comes from a bootstrap with 2000 resamples. For the classification metrics, both y and p are mapped to the density levels Low (<50 people), Medium (50-149), High (150-499) and Very high (≥500); accuracy, precision, recall, F1 and the confusion matrix are then computed on these levels. Because the levels are ordered, the quadratic-weighted Cohen's kappa is also reported.

3. Training behaviour

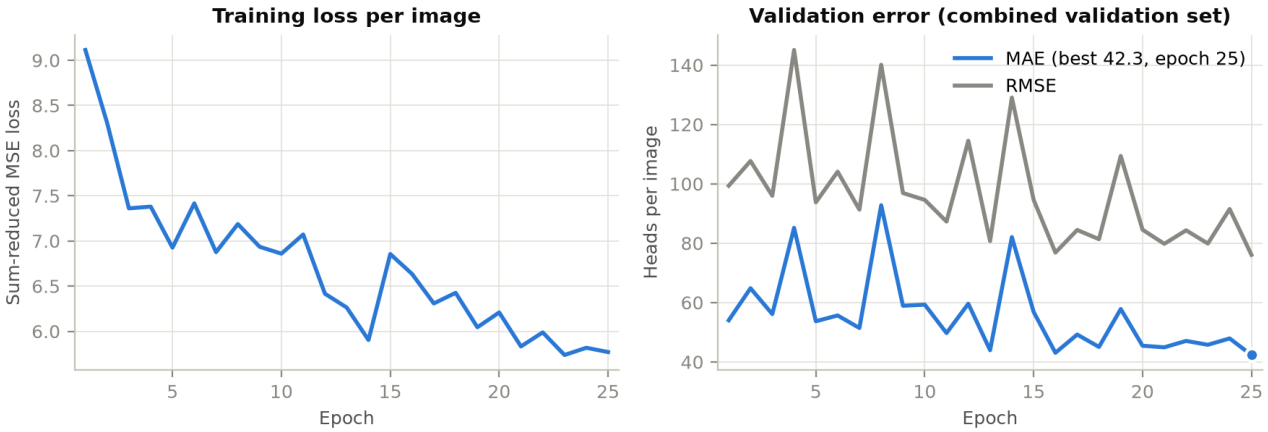


Figure 1. Training loss and validation error of the joint fine-tuning run. The marked point is the checkpoint evaluated in this report.

4. Counting results

Table 3. Counting performance of the final model on the held-out test splits. r is the Pearson correlation; "±20%" is the share of images whose predicted count is within 20% of the true count.

Test set	Images	MAE	MAE 95% CI	RMSE	Bias	R^2	r	Count acc.	±20%
ShanghaiTech A	182	77.25	63.2 - 92.5	126.91	-17.4	0.871	0.941	81.2%	60.4%
ShanghaiTech B	316	19.11	16.8 - 21.8	29.63	+3.3	0.903	0.957	78.2%	63.9%
Roboflow	458	48.50	42.3 - 55.3	86.05	+1.1	0.810	0.902	67.2%	42.4%
All test images	**956**	**44.26**	**39.9 - 48.9**	**83.09**	**-1.7**	**0.881**	**0.942**	**73.5%**	**52.9%**

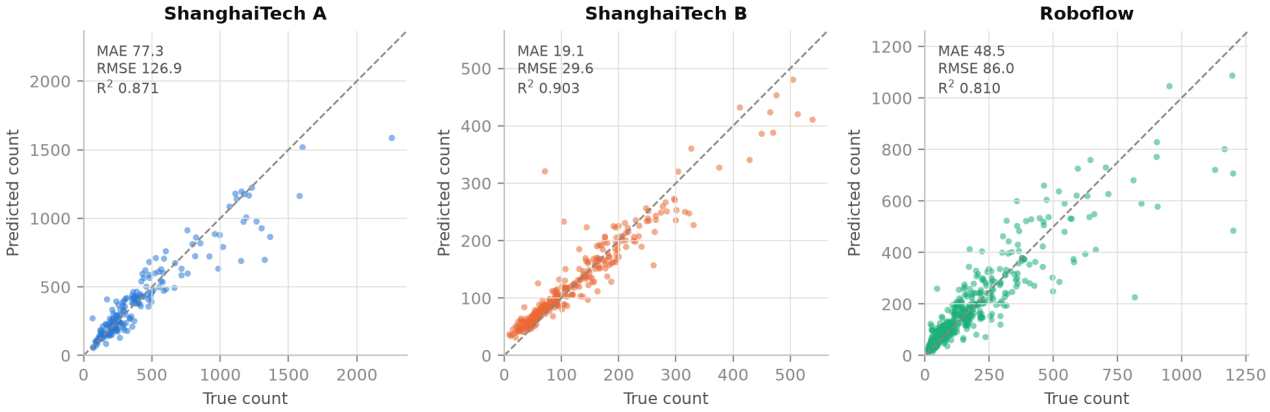


Figure 2. Predicted versus true count for every test image. Points on the dashed line are perfect predictions; points above it are over-counts.

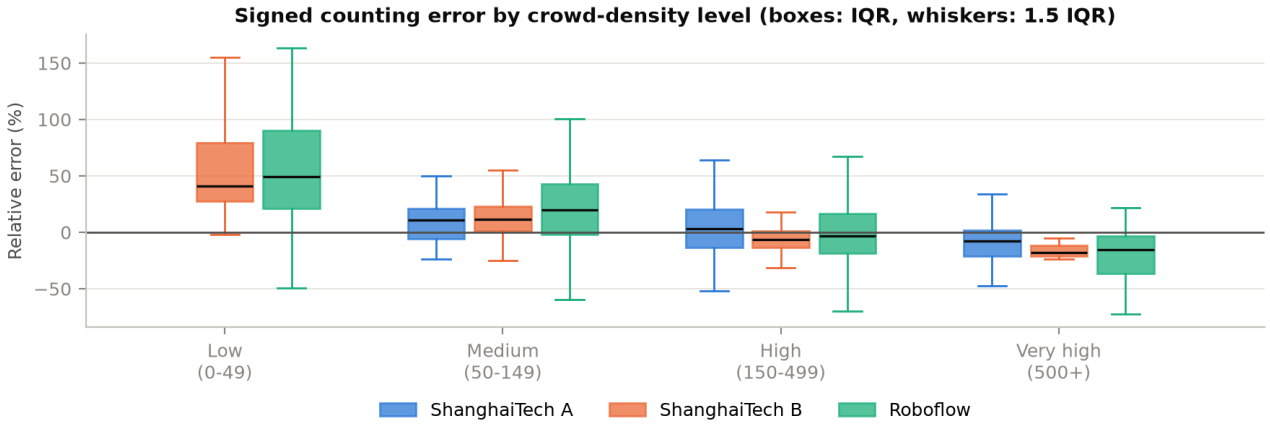


Figure 3. Distribution of the signed relative error (predicted - true) / true, split by the true crowd-density level. Levels with fewer than three images in a dataset are omitted.

The error depends on crowd density. Over all test images, the median relative error is +48% for Low, +15% for Medium, -3% for High, -11% for Very high scenes: the model tends to over-count sparse scenes and under-count the densest ones, the typical regression-to-the-mean behaviour of density-map counters.

Table 4. Comparison with published results on the ShanghaiTech test sets. Published numbers are taken from the cited papers, which train a separate model per part; this work uses one model for all datasets.

Method	Source	Part A MAE	Part A RMSE	Part B MAE	Part B RMSE
MCNN	Zhang et al., CVPR 2016	110.2	173.2	26.4	41.3
CSRNet	Li et al., CVPR 2018	68.2	115.0	10.6	16.0
This work (joint model)	**this report**	**77.3**	**126.9**	**19.1**	**29.6**

Compared with the published single-dataset CSRNet, the joint model's MAE is +9.1 on Part A and +8.5 on Part B. Sharing one set of weights across datasets with very different densities, image sizes and annotation styles trades some per-dataset accuracy for a single model that covers sparse street scenes, dense crowds and the Roboflow images at once.

5. Crowd-density level classification

A monitoring system acts on the crowd level rather than on the exact count, so the counts were also evaluated as a four-class problem. Each test image is assigned a true level from its annotated count and a predicted level from the model's count; Figure 4 shows the resulting confusion matrix over all 956 test images.

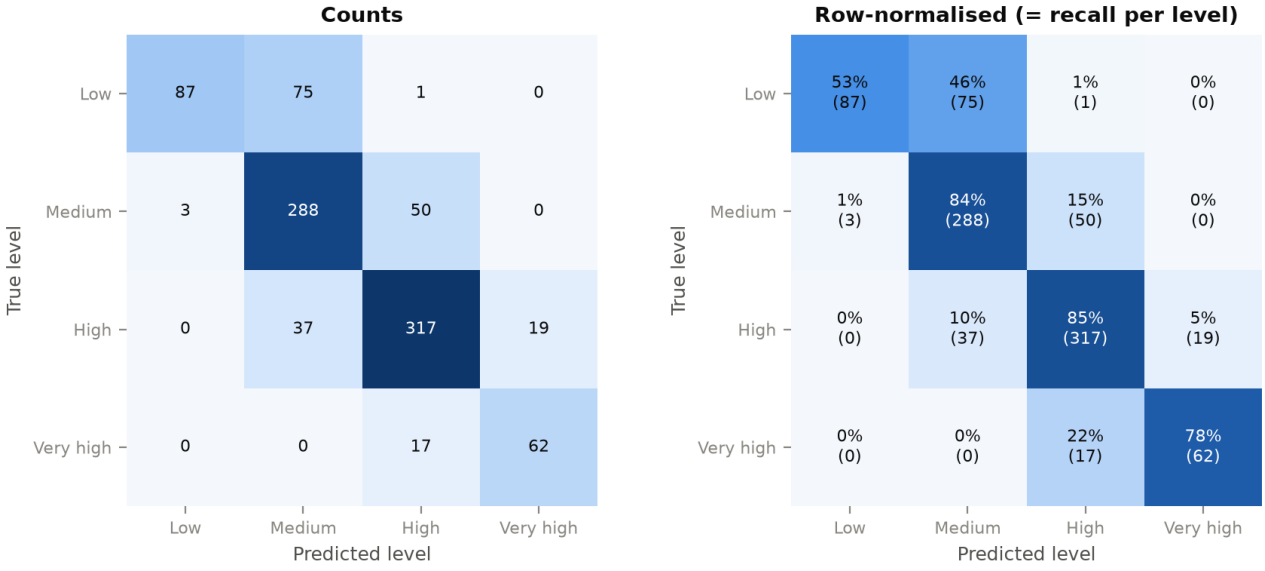


Figure 4. Confusion matrix of the crowd-density levels over all test images: absolute counts (left) and normalised per true level (right). The diagonal holds the correct predictions.

99.5% of the misclassified images land in a neighbouring level. The most frequent confusion is true Low predicted as Medium (75 images, 46% of the Low images), consistent with the over-counting of sparse scenes seen in Figure 3.

Table 5. Per-level precision, recall and F1 over all test images.

Level	People	Images	Precision	Recall	F1 score
Low	0-49	163	96.7%	53.4%	68.8%
Medium	50-149	341	72.0%	84.5%	77.7%
High	150-499	373	82.3%	85.0%	83.6%
Very high	500+	79	76.5%	78.5%	77.5%
Macro average		**956**	**81.9%**	**75.3%**	**76.9%**

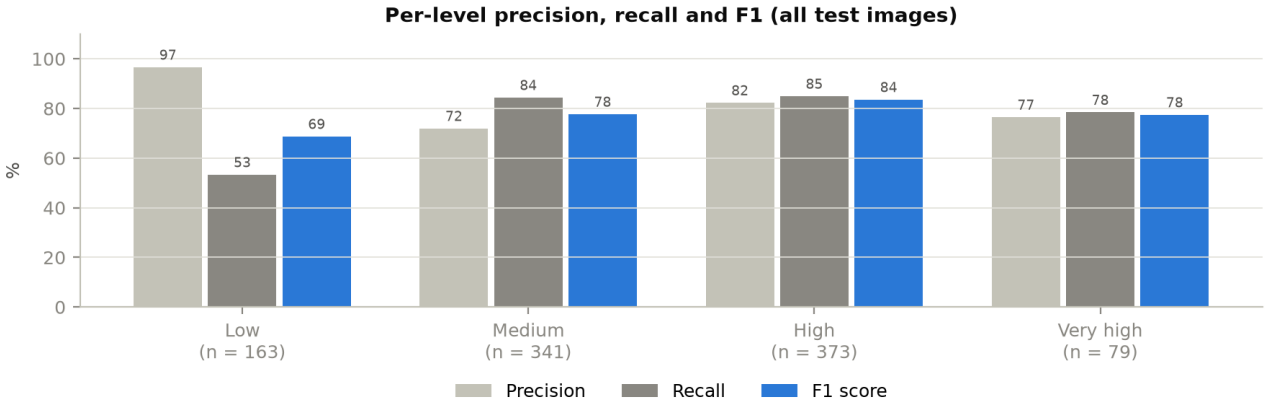


Figure 5. Per-level precision, recall and F1 score (all test images).

Table 6. Density-level classification metrics per test set. Macro F1 averages the levels that occur in that test set's ground truth or predictions.

Test set	Images	Accuracy	Within one level	Macro F1	Weighted F1	Weighted kappa
ShanghaiTech A	182	85.7%	100.0%	81.7%	85.6%	0.798
ShanghaiTech B	316	82.9%	100.0%	58.6%	81.0%	0.801
Roboflow	458	73.4%	99.8%	72.7%	73.5%	0.812
All test images	**956**	**78.9%**	**99.9%**	**76.9%**	**78.5%**	**0.842**

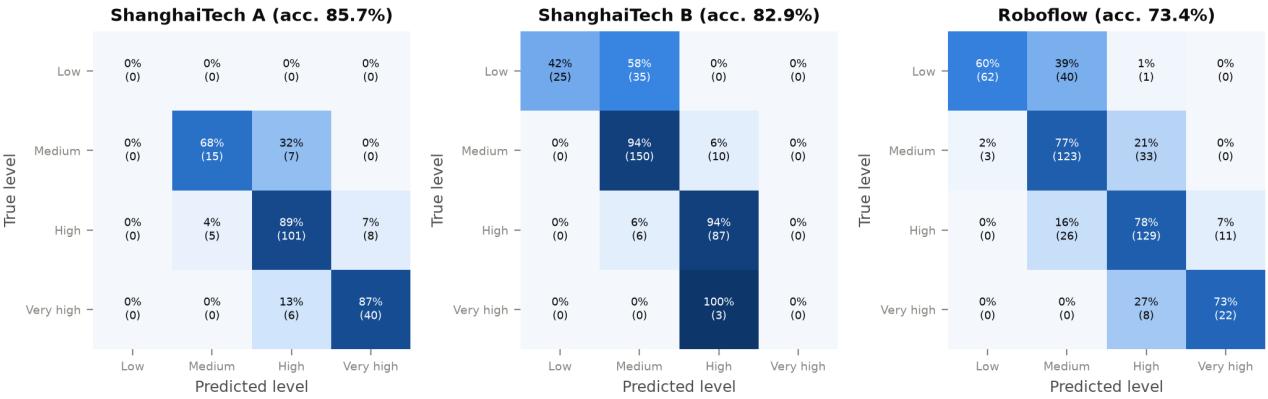


Figure 6. Row-normalised confusion matrices per test set.

6. Over-crowding alert detection

The deployed system raises an alert when the estimated count of a zone reaches a threshold. Treating "true count ≥ T" as the positive class, Figure 7 shows the F1 score of this binary decision for every threshold T (only thresholds with at least five positive images in a dataset are plotted).

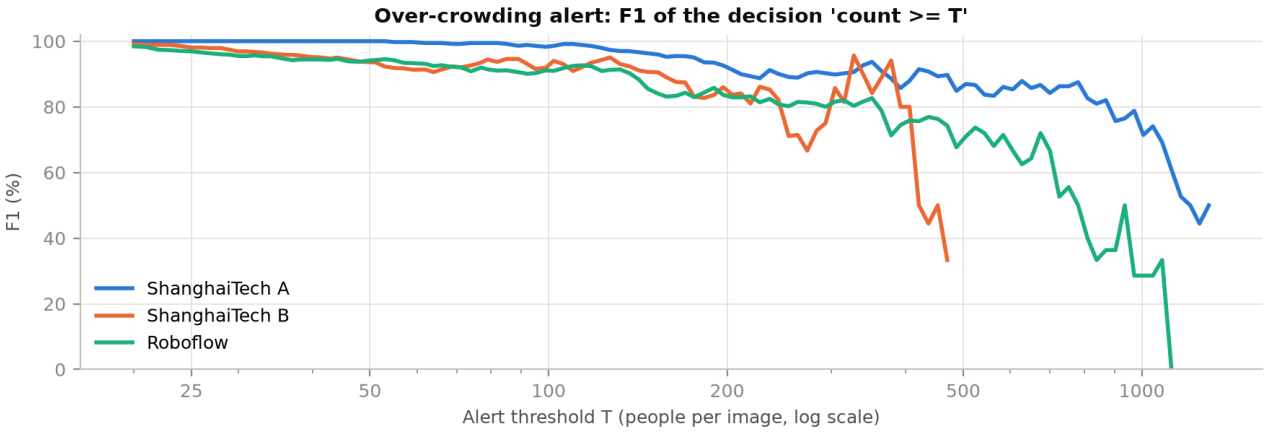


Figure 7. F1 score of the over-crowding alert as a function of the alert threshold T.

Table 7. Alert detection at selected thresholds over all test images.

Threshold T	Positive images	Precision	Recall	F1 score
50	793	91.2%	99.6%	95.2%
100	579	90.3%	96.5%	93.3%
150	452	89.1%	91.8%	90.4%
250	248	85.4%	82.3%	83.8%
500	79	76.5%	78.5%	77.5%

7. Qualitative results

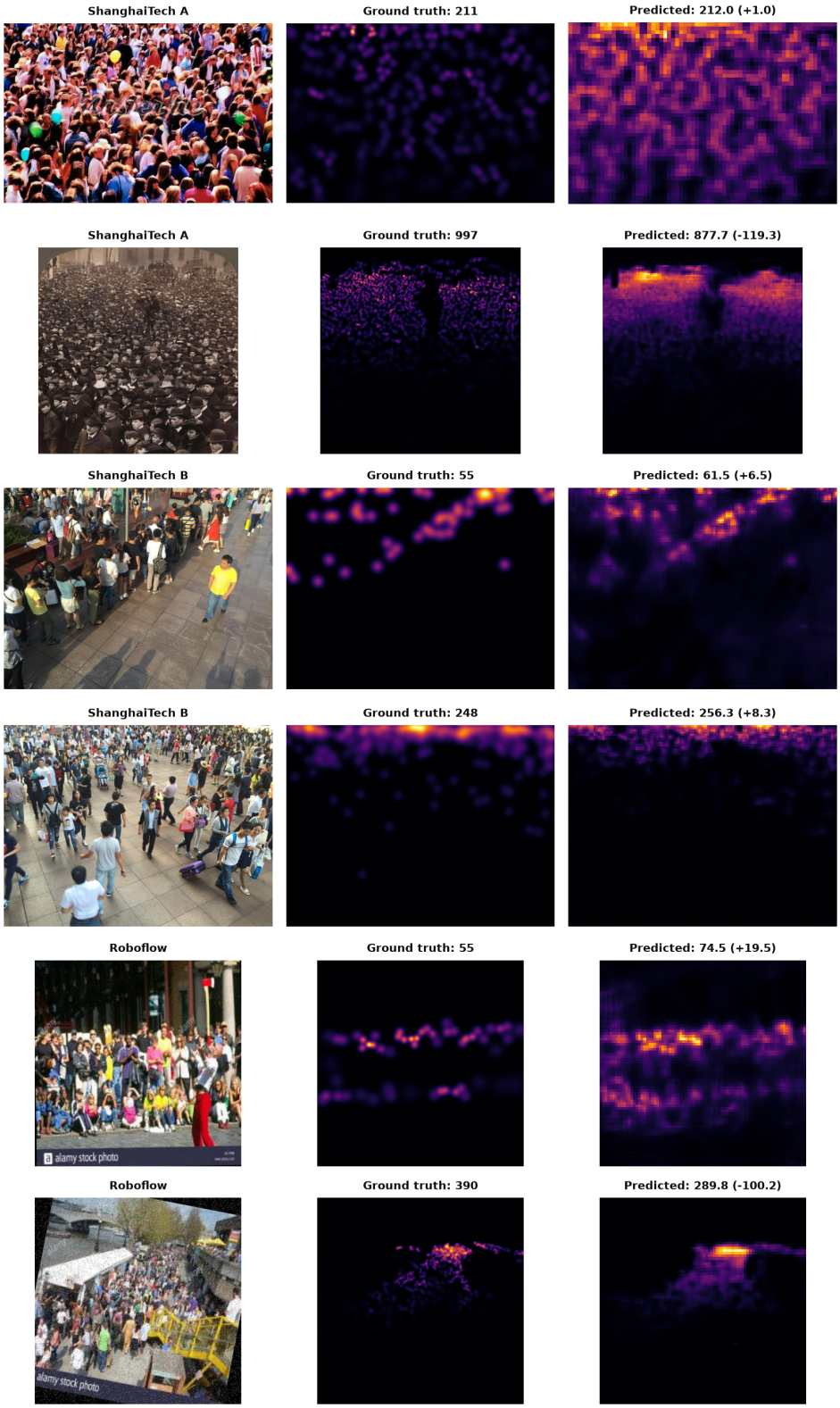


Figure 8. Test images (left), ground-truth density maps (middle) and predicted density maps (right). For each dataset, one image at the 25th and one at the 90th percentile of the true count is shown.

8. Inference speed

Table 8. CPU inference time measured during evaluation (Intel64 Family 6 Model 140 Stepping 2, GenuineIntel, 4 threads, PyTorch 2.14.1+cpu, float32, batch 1).

Test set	Image size	Median time / image	Images / second
ShanghaiTech A	0.59 MP (median)	4.15 s	0.24
ShanghaiTech B	0.79 MP (median)	4.92 s	0.20
Roboflow	0.41 MP (median)	2.59 s	0.39

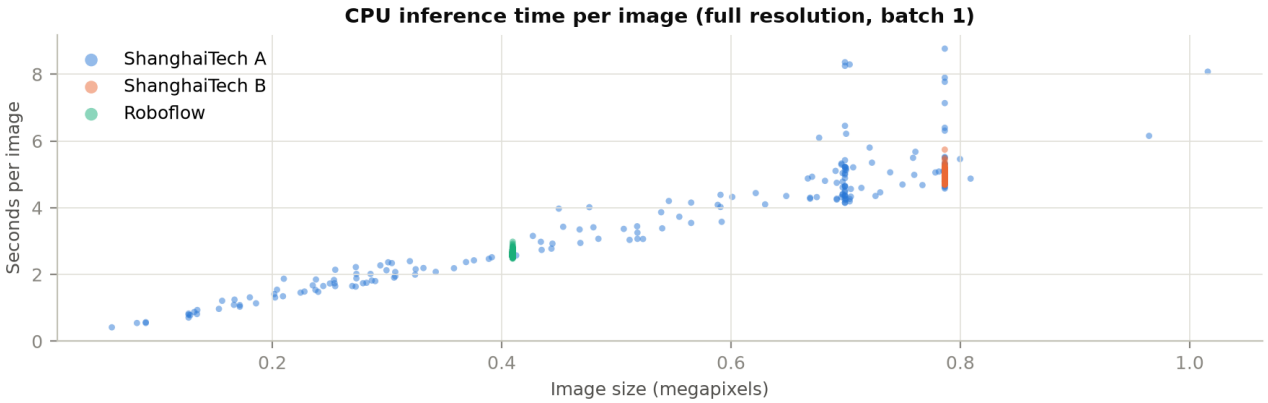


Figure 9. CPU inference time per test image against image size.

9. Notes on the evaluation

• All numbers in this report are computed by research/make_results_pdf.py from the model's predictions on the test images; only the published baselines in Table 4 are taken from the literature.

• The ShanghaiTech test sets were also used to choose the best training epoch, which is the usual practice in crowd-counting papers but makes their scores slightly optimistic. Roboflow model selection used the separate validation split, so the Roboflow test score is fully held out.

• The confusion matrix, accuracy and F1 score are not native to crowd counting; they depend on the chosen level boundaries (50, 150 and 500 people per image), which were fixed from the ground-truth count distribution before the model's predictions were examined.

• The phone photographs are not part of any evaluation because they have no human annotations.